

25CY201 SDG No. 4,9,17	APPLIED STATISTICS USING PYTHON (Common to all Computing Sciences)	L	T	P	C
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COURSE OBJECTIVES

1. To develop a strong understanding of modern computational statistical approaches and their applications to various datasets.
2. To gain a comprehensive understanding of linear regression models.
3. To learn how to test hypotheses using statistical analysis.
4. To understand key technologies in data science and business analytics.
5. To effectively visualize data using tools like Matplotlib and Seaborn to communicate statistical findings.

INTRODUCTION TO PYTHON AND DESCRIPTIVE STATISTICS

9

Introduction to basics of Python, NumPy and SciPy - Basic data handling using Pandas - Plotting using Matplotlib (Histogram, Scatter, Bar, Box and Line plot).

Measures of Central Tendency: Mean, Median, Mode - Measures of Dispersion: Range, Quartile Deviation and Standard Deviation - Summarization of univariate and bivariate data.

CORRELATION, REGRESSION AND PREDICTIVE ANALYSIS

9

Correlation: Karl Pearson's correlation and Spearman's Rank correlation - Linear Regression: Interpretation of regression coefficients - Standard multiple regression models.

Correlation computation using Python libraries - Introduction to Scikit-learn, implementation of linear regression models - Visualization of regression lines.

TESTING OF HYPOTHESIS AND ANALYSIS OF VARIANCE

9

Statistical hypothesis: Type I and Type II errors –Large samples: Difference of Means and proportions - Small samples: t-test, F-test, Chi-square test - Analysis of variance: One-way and Two-way.

Hypothesis testing using SciPy - Analysis of variance: Implementation -Interpretation of p-values - visualization of test results.

PRINCIPAL COMPONENT ANALYSIS AND DATA AGGREGATION

9

Eigenvalues and Eigenvectors of real symmetric matrices - Principal Component Analysis: Principal components - Algorithm for conducting principal component analysis - Analytical Approach within 3x3 matrix - H-plot.

Group by Mechanics - Data Aggregation - Group wise Operations and Transformations - Pivot Tables and Cross Tabulations - Principal Component Analysis using Python.

TIME SERIES AND PATTERN ANALYSIS

9

Time Series analysis: Data Ranges - Frequencies and Shifting Visualization in Python - Advanced visualization using Matplotlib and Seaborn.

Introduction to Pattern Analysis – Types of clustering – Pattern grouping using partition-based methods – Centroid-based clustering (K-Means): Euclidean distance – Hierarchy-based clustering: Agglomerative and Divisive approaches.

TOTAL PERIODS (Theory): 45

LIST OF EXPERIMENTS

1. Execution of matrix operations using NumPy to perform in Data analysis.
2. Computation of mean, variance and standard deviation using Python.
3. Data visualization for univariate and bivariate datasets.
4. Correlation and regression analysis using Python.
5. Hypothesis testing using t-test and Chi-square test.
6. Analysis of variance - based comparison of datasets.
7. Principal Component Analysis using Python.
8. K-Means clustering for engineering datasets.
9. Time-series trend analysis.
10. Mini project: Statistical analysis of real-world data.

*Note: Data Sets: Students Performance Data, Medical Data, HR Data, Agriculture Data, Financial Data

TOTAL PERIODS (Lab): 30

TOTAL: 75 PERIODS

COURSE OUTCOMES

Upon completion of the course, students shall have ability to

CO1	Understand the measures of central tendency and dispersion to create appropriate data visualizations.	[U]
CO2	Visualize data relationships and evaluate model performance through correlation and regression.	[AP]
CO3	Analyze the concept of hypothesis testing for small and large samples using various tests.	[AN]
CO4	Apply statistical methods to identify principal components and aggregate data using various methods.	[AP]
CO5	Analyze different clustering algorithms for real world data.	[AN]

TEXTBOOKS

1. Wes Mc Kinney, "Python for Data Analysis", O'Reilly, 3rd Edition 2022.
2. Gupta, S.C. and Kapoor, V.K, Fundamentals of Mathematical Statistics, 12th Edition, Sultan Chand & Sons, 2020.
3. Jobson. J.D, "Applied Multivariate Data Analysis", Vol II, Springer, 2012.

REFERENCE BOOKS

1. Miller. I.R, Freund. J.E and Johnson. R, "Probability and Statistics for Engineers, Pearson Education, Asia, 9th Edition, 2016.
2. Brian S. Everitt, Graham Dunn, "Applied Multivariate Data Analysis" John Wiley & Sons, Ltd, 2nd Edition, 2001.
3. Mark Summerfield, Programming in Python 3: A Complete introduction to the Python Language, Addison-Wesley Professional, 2009.

WEB RESOURCES

1. <https://www.kdnuggets.com/2016/07/statistical-data-analysis-python.html>
2. <https://people.duke.edu/~ccc14/sta-663/>
3. <http://greenteapress.com/wp/think-python/>